

Question One: (15 Marks) *Start a new sheet of paper.*

a) Find $\int \cos^3 x dx$. [2]

b) Find $\int \frac{dx}{\sqrt{x^2 + 2x + 5}}$. [2]

c) Evaluate $\int_1^2 x \ln x dx$ (in exact form). [3]

d)

i) Find real numbers a, b and c such that

$$\frac{1}{(x^2 + 1)(x + 1)} = \frac{ax + b}{(x^2 + 1)} + \frac{c}{(x + 1)}.$$
 [2]

ii) Hence evaluate $\int_0^1 \frac{1}{(x^2 + 1)(x + 1)} dx$ (in exact form). [2]

e) Use the substitution $x = \sin^2 \theta$ to evaluate $\int_0^{\frac{1}{2}} \frac{\sqrt{x}}{(1-x)^{\frac{3}{2}}} dx$ [4]

Question Two: (15 Marks) *Start a new sheet of paper.*

a) Given that $z = \sqrt{3} + \frac{1+i}{1-i}$ find:

i) $\text{Im}(z)$ [1]

ii) $\bar{z}$ [1]

iii) z in mod/arg form. [2]

b) Solve $z^2 = 3 - 4i$. [2]

c) Illustrate on an Argand diagram the region given by

$$\left\{ z : 0 \leq \arg(z + 4 + i) \leq \frac{2\pi}{3} \text{ and } |z + 4 + i| \leq 4 \right\}.$$
 [3]

(Question 2 continued over)

d) z is a point on the circle $|z - 1| = 1$ and $\arg(z) = \theta$.

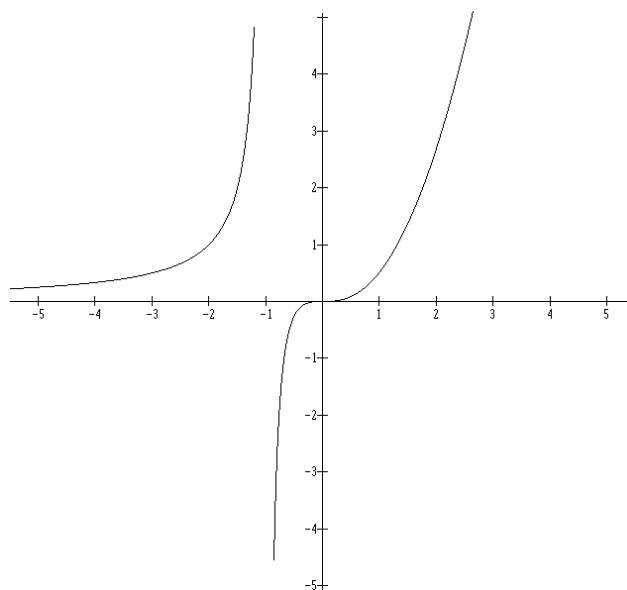
i) Find $\arg(z - 1)$ in terms of θ . [1]

ii) Hence find $\arg(z^2 - 3z + 2)$ in terms of θ . [2]

e) Find the complex fifth root of $-i$, in mod/arg form, and show these roots on an Argand diagram. [3]

Question Three: (15 Marks) *Start a new sheet of paper.*

a) The diagram shows the graph of $y = F(x)$. Draw neat sketches of (each should take about one third of a page):



i) $y = \frac{1}{F(x)}$ [2]

ii) $y = F(x) - |F(x)|$ [2]

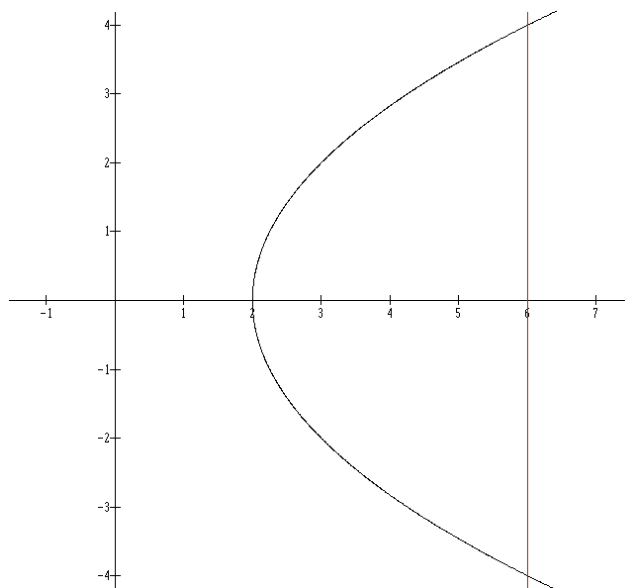
iii) $y = x.F(x)$ [2]

iv) $y = e^{F(x)}$ [2]

v) $y = \sqrt{F(x)}$ [2]

(Question 3 continued over)

- b) The diagram shows the region bounded by the curve $y^2 = 4(x - 2)$ and the line $x = 6$. Use the method of cylindrical shells to find the volume of the solid formed by rotating the given region about the y -axis. [5]



Question Four: (15 Marks) *Start a new sheet of paper.*

- a)
- Derive the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$. [2]
 - If $P(a \cos \theta, b \sin \theta)$ is on the ellipse in the first quadrant, and the tangent at P meets the x -axis and the y -axis at X and Y respectively, find the coordinates of X and Y . [2]
 - For the triangle thus formed by OXY , find the minimum area of this triangle, and the coordinates of P (in terms of a and b) for this case. [5]
- b)
- Given $I_n = \int_0^{\frac{\pi}{4}} \tan^n x dx$ where n is a positive integer and $n \geq 2$, show that $I_n = \frac{1}{n-1} - I_{n-2}$. [4]
 - Hence evaluate $I_5 = \int_0^{\frac{\pi}{4}} \tan^5 x dx$. [2]

Question Five: (15 Marks) *Start a new sheet of paper.*

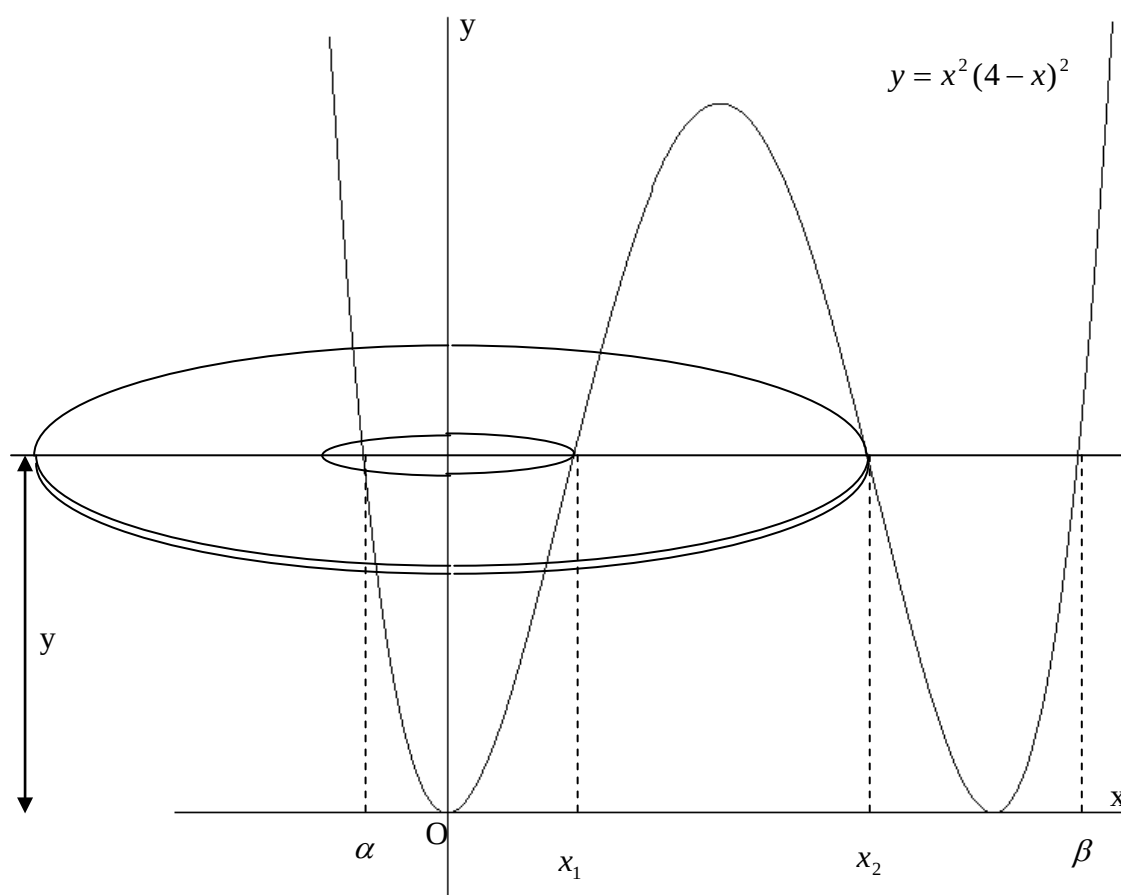
- a) Factorise $Q(x) = x^6 - 3x^2 + 2$ over the complex number field, given that it has two double roots. [3]
- b) The equation $x^3 + px + 1 = 0$ has three real non-zero roots α, β and δ .
- i) Find the values of $\alpha^2 + \beta^2 + \delta^2$ and $\alpha^4 + \beta^4 + \delta^4$ in terms of p , and show that p must be strictly negative. [4]
- ii) Find the monic equation, with co-efficients in terms of p , whose roots are $\frac{\alpha}{\beta\delta}, \frac{\beta}{\alpha\delta}, \frac{\delta}{\alpha\beta}$. [4]
- c) Let z_1, z_2 and z_3 be three complex numbers represented by the points Z_1, Z_2 and Z_3 respectively on the Argand diagram, where $z_1 \times z_3 = (z_2)^2$. Show that OZ_2 bisects $\angle Z_1OZ_3$. [4]

Question Six: (15 Marks) *Start a new sheet of paper.*

- a) PQRS is a cyclic quadrilateral. The bisector of $\angle PQS$ cuts the segment PR at X and the circle at M, and RM cuts the segment QS at Y.
- i) Draw a neat diagram showing the above information. [1]
- ii) Prove XQRY is a cyclic quadrilateral. [3]
- iii) Prove XY is parallel to PS. [3]
- b) Find the limiting sum of the series $\frac{1}{5} + \frac{2}{5^2} + \frac{3}{5^3} + \dots + \frac{n}{5^n} + \dots$ [3]
- c) From DeMoivre's Theorem, we know $\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$. Use this to solve the equation $16x^4 - 16x^2 + 1 = 0$, and deduce the exact values of $\cos \frac{\pi}{12}$ and $\cos \frac{5\pi}{12}$. [5]

Question Seven: (15 Marks) *Start a new sheet of paper.*

a)



The region between $x = 0$ and $x = 4$ is rotated about the y -axis. The volume of the solid formed is found by taking slices perpendicular to the y -axis. The typical slice shown in the diagram is at a height y above the x -axis.

- i) Deduce that α, x_1, x_2 and β , as shown in the diagram, are the roots of $x^4 - 8x^3 + 16x^2 - y = 0$. [1]
- ii) Use the symmetry in the graph to explain why $\frac{x_1 + x_2}{2} = 2$ and $\frac{\alpha + \beta}{2} = 2$. Hence, by considering the co-efficients of the equation in (i), show that $\alpha\beta = -x_1x_2$, and deduce that $x_1x_2 = \sqrt{y}$ and that $x_2 - x_1 = 2\sqrt{4 - \sqrt{y}}$. [5]

(Question 8(a) continued over)

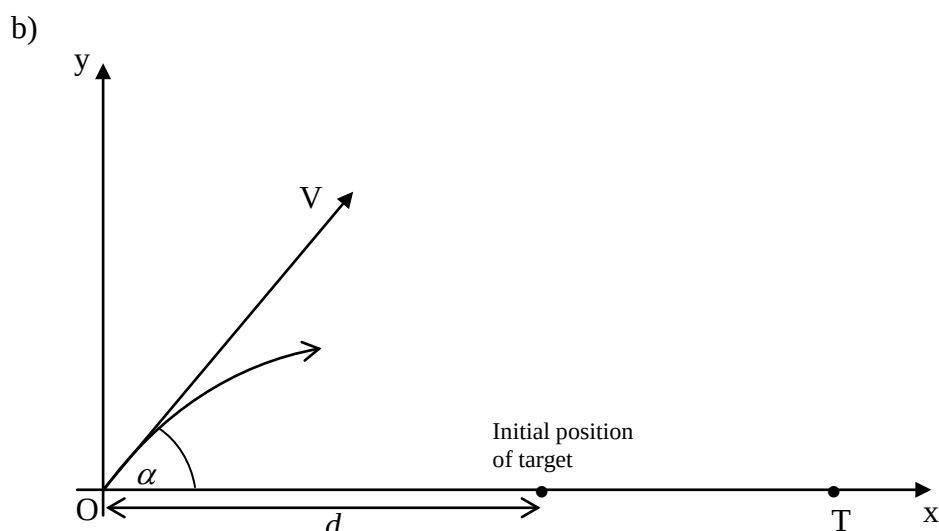
iii) Show that the volume of the solid of revolution is given by

$$V = 8\pi \int_0^{16} \sqrt{4 - \sqrt{y}} dy.$$

Use the substitution $y = (4 - u)^2$ to evaluate

this integral and find the exact volume.

[4]



A projectile, of initial speed V m/s, is fired at an angle α from the origin O towards a target T , which is moving away from O along the x -axis.

You may assume that the projectile's trajectory is defined by the equations:

$x = Vt \cos \alpha$ and $y = Vt \sin \alpha - \frac{1}{2}gt^2$, where x and y are the horizontal and vertical displacements of the projectile in meters at time t seconds after firing, and where g is the acceleration due to gravity.

i) Show that the projectile is above the x -axis for a total of $\frac{2V \sin \alpha}{g}$ seconds.

[1]

ii) Show that the horizontal range of the projectile is $\frac{2V^2 \sin \alpha \cos \alpha}{g}$ meters.

[1]

iii) At the instant the projectile is fired, the target T is d meters from O and is moving away at a constant speed of u m/s.

Suppose that the projectile hits the target when fired at an angle of elevation α . Show that $u = V \cos \alpha - \frac{gd}{2V \sin \alpha}$.

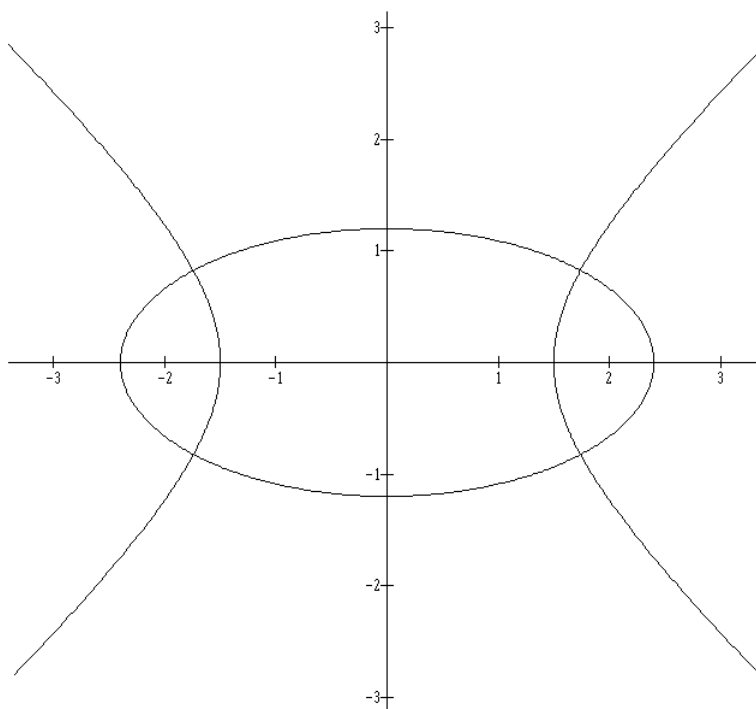
[3]

Question Eight: (15 Marks) *Start a new sheet of paper.*

- a) Find the volume of the solid generated by rotating the region common to the circles $x^2 + y^2 = 16$ and $x^2 + y^2 = 8x$ about their common chord.

[8]

- b) Hyperbola H has equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and eccentricity e . Ellipse E has equation $\frac{x^2}{(a^2 + b^2)} + \frac{y^2}{b^2} = 1$. See diagram below.



- i) Show that ellipse E has eccentricity $\frac{1}{e}$.

[1]

- ii) If H and E intersect at P in the first quadrant, show that the acute angle α between the tangents to H and E at P is given by $\tan \alpha = \sqrt{2} \left(e + \frac{1}{e} \right)$.

[6]

$$\begin{aligned} 1) a) \int \cos^3 x \, dx &= \int \cos x (1 - \sin^2 x) \, dx \\ &= \int \cos x - \sin^2 x \cos x \, dx \\ &= \sin x - \frac{1}{3} \sin^3 x + C \end{aligned}$$

$$\begin{aligned} b) \int \frac{dx}{\sqrt{x^2 + 2x + 5}} &= \int \frac{dx}{\sqrt{x^2 + 2x + 1 + 4}} \\ &= \int \frac{dx}{\sqrt{(x+1)^2 + 2^2}} \\ &= \ln \left((x+1)^2 + \sqrt{(x+1)^2 + 4} \right) + C \end{aligned}$$

$$\begin{aligned} d) i) \frac{1}{(x^2+1)(x+1)} &= \frac{Ax+B}{x^2+1} + \frac{C}{x+1} \\ \Rightarrow 1 &= (x+1)(Ax+B) + C(x^2+1) \\ x = -1 \text{ gives } 1 &= 2C \quad \text{ie } C = \frac{1}{2} \\ x = 0 \text{ gives } 1 &= B+C \quad \text{ie } B = \frac{1}{2} \\ x = 1 \text{ gives } 1 &= 2(A+B) + 2C \quad \text{ie } A = -\frac{1}{2} \\ \therefore \frac{1}{(x^2+1)(x+1)} &= \frac{1-x}{2(x^2+1)} + \frac{1}{2(x+1)} \end{aligned}$$

$$\begin{aligned} ii) \int_0^1 \frac{1}{(x^2+1)(x+1)} \, dx &= \frac{1}{2} \int_0^1 \frac{1-x}{(x^2+1)} + \frac{1}{x+1} \, dx \\ &= \frac{1}{2} \int_0^1 \frac{1}{x^2+1} + \frac{-x}{x^2+1} + \frac{1}{x+1} \, dx \\ &= \frac{1}{2} \left[\tan^{-1} x - \frac{1}{2} \ln(x^2+1) + \ln(x+1) \right]_0^1 \\ &= \frac{1}{2} \left[\tan^{-1} 1 - \frac{1}{2} \ln 2 + \ln 2 - \left(\tan^{-1} 0 - \frac{1}{2} \ln 1 + \ln 1 \right) \right] \\ &= \frac{1}{2} \left(\frac{\pi}{4} + \frac{1}{2} \ln 2 - 0 \right) \\ &= \frac{\pi + 2 \ln 2}{8} \end{aligned}$$

Marking Guidelines

① appropriate substitution

① correct soln inc. "+C"

① correct manipulation of root to a standard form

① correct use of standard integral

① forms correct equivalent equations

① correct values of a, b, c

① correct integrations

① answer correct

Comments

well done

• some students had difficulty once the root was in a standard form

well done

well done

$$\begin{aligned} Q1) d) \int_1^2 x \ln x \, dx & \quad u = \ln x \quad dv = x \\ & \quad du = \frac{1}{x} \quad v = \frac{1}{2} x^2 \\ \therefore &= \left[\frac{1}{2} x^2 \ln x \right]_1^2 - \int_1^2 \frac{1}{2} x^2 \cdot \frac{1}{x} \, dx \\ &= \left[\frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 \right]_1^2 \\ &= \left(2 \ln 2 - 1 - \left(0 - \frac{1}{4} \right) \right) \\ &= 2 \ln 2 - \frac{3}{4} \\ &= \frac{4 \ln 2 - 3}{4} \end{aligned}$$

$$\begin{aligned} e) \int_0^{\frac{1}{2}} \frac{\sqrt{x} \, dx}{(1-x)^{3/2}} & \quad x = \sin^2 \theta \\ & \quad dx = 2 \sin \theta \cdot \cos \theta \, d\theta \\ & \quad x = \frac{1}{2} \quad \sin^2 \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{4} \\ & \quad x = 0 \quad \sin^2 \theta = 0 \Rightarrow \theta = 0 \\ \therefore &= \int_0^{\frac{\pi}{4}} \frac{\sqrt{\sin^2 \theta} \cdot 2 \sin \theta \cos \theta \, d\theta}{(1 - \sin^2 \theta)^{3/2}} \\ &= 2 \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta \cos \theta}{\cos^3 \theta} \, d\theta \\ &= 2 \int_0^{\frac{\pi}{4}} \tan^2 \theta \, d\theta \\ &= 2 \int_0^{\frac{\pi}{4}} (\sec^2 \theta - 1) \, d\theta \\ &= 2 \left[\tan \theta - \theta \right]_0^{\frac{\pi}{4}} \\ &= 2 \left(1 - \frac{\pi}{4} - (0) \right) \\ &= 2 - \frac{\pi}{2} \end{aligned}$$

Marking Guidelines

① correct play parts u, v

① correct definite integral method

① correct answer

① correct change of limits

① resolves to $\tan^2$ ① correct $\tan^2$ integration

① correct answer

Comments

well done

• well done apart from careless errors such as omitting 2.

$$\begin{aligned} \text{Q2) a) } z &= \sqrt{3} + \frac{1-i}{1-i} \times \frac{(1+i)}{(1+i)} \\ &= \sqrt{3} + \frac{2i}{2} \\ &= \sqrt{3} + i \end{aligned}$$

$$\therefore \text{(i) } \operatorname{Re}(z) = 1$$

$$\text{(ii) } \bar{z} = \sqrt{3} - i$$

$$\begin{aligned} \text{(iii) } |z| &= \sqrt{a^2 + b^2} \quad \arg(z) = \tan^{-1} \frac{b}{a} \\ &= \sqrt{3+1} \quad = \frac{\pi}{6} \\ &= 2 \end{aligned}$$

$$\therefore z = 2 \cos \frac{\pi}{6}$$

$$\text{b) } z^2 = 3 - 4i$$

$$\text{a) } (a+ib)^2 = 3-4i \quad a, b \text{ real}$$

$$a^2 - b^2 + 2abi = 3 - 4i$$

$$\therefore a^2 - b^2 = 3 \quad 2ab = -4$$

$$b = \frac{-2}{a} \quad \text{in } \textcircled{1}$$

$$\therefore a^2 - \left(\frac{-2}{a}\right)^2 = 3$$

$$a^2 - \frac{4}{a^2} = 3$$

$$\therefore a^4 - 3a^2 - 4 = 0$$

$$(a^2 - 4)(a^2 + 1) = 0$$

$$\text{so } a^2 = 4 \text{ or } a^2 = -1 \text{ (reject as } a \text{ is real)}$$

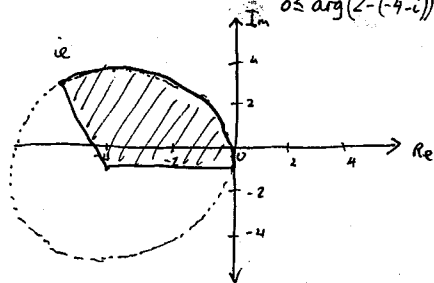
$$\therefore a = \pm 2 \quad b = \mp 1$$

$$\therefore z_1 = 2 - i \quad z_2 = -2 + i //$$

$$\begin{aligned} \text{c) } z + 4 + i &\Rightarrow |z - (-4-i)| \leq 4, \text{ a circle} \\ &= z - (-4-i) \end{aligned}$$

$$\text{centre } (-4, -1)$$

$$0 \leq \arg(z - (-4-i)) \leq \frac{2\pi}{3} \Rightarrow \text{sector.}$$



Marking Guidelines

Comments

①

①

① mod

① angle

① appropriate method for getting $\sqrt{}$ ① correct solutions for z_1, z_2

① correct | | interpret.

① correct arg interpret.

① correct diagram.

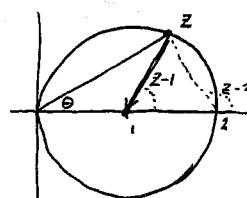
• some wished to include the "i" in $\operatorname{Im}(z)$.

• some did mod/arg of $\bar{z}$.

• remember to write down solutions, not just state a and b!

• some placed the center not at $(-4, -1)$.

d)



$$\text{i) } \arg(z-1) = 2\theta$$

$$\begin{aligned} \text{ii) } \arg(z^2 - 3z + 2) &= \arg[(z-2)(z-1)] \\ &= \arg(z-2) + \arg(z-1) \end{aligned}$$

$$\arg(z-2) \text{ is external angle of } \Delta, \text{ so}$$

$$\arg(z-2) = \frac{\pi}{2} + \theta$$

$$\begin{aligned} \therefore \arg(z^2 - 3z + 2) &= \arg(z-2) + \arg(z-1) \\ &= \frac{\pi}{2} + \theta + 2\theta \\ &= 3\theta + \frac{\pi}{2} \end{aligned}$$

$$\text{e) } -i = \cos(-\frac{\pi}{2})$$

$$\text{let } z = r \cos \theta$$

$$\therefore z^5 = r^5 \cos 5\theta \quad (\text{by De Moivre's Theorem})$$

$$\therefore r^5 \cos 5\theta = \cos(-\frac{\pi}{2})$$

$$\therefore r = 1, \quad 5\theta = 2k\pi - \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{10}(4k-1) \quad k=0, 1, 2, 3, 4.$$

$$\therefore z_0 = \cos(-\frac{\pi}{10}) (= \cos \frac{19\pi}{10})$$

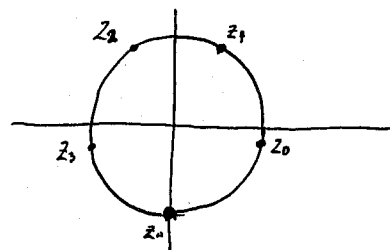
$$z_1 = \cos(\frac{3\pi}{10})$$

$$z_2 = \cos(\frac{7\pi}{10})$$

$$z_3 = \cos(\frac{11\pi}{10})$$

$$z_4 = \cos(\frac{15\pi}{10})$$

$$= \cos(\frac{3\pi}{2}) (= -i)$$



Marking Guidelines

Comments

①

① correct arg(z-2)

① correct answer

① correct use of De Moivre and general solution.

① correct roots listed

① diagram correct

• those who drew a diagram did quite well. Those that did not, very poorly on this question!

• many tried to use π - the straight line - unsuccessfully.

• many had wrong $\cos(-i)$!

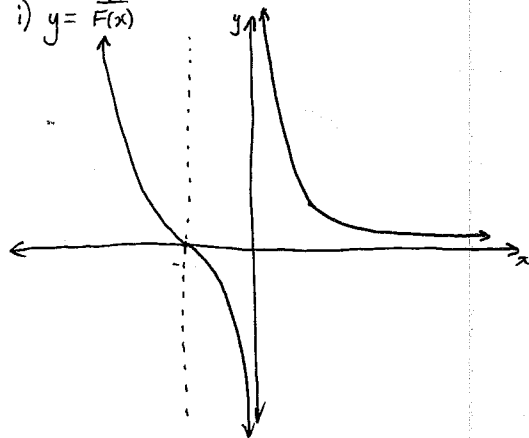
• many did not make clear connections in the working, between DMT and eqn.

• general soln poorly done

• many incorrect due to not graphing $-i$ correctly!

3) a)

i) $y = \frac{1}{F(x)}$



Marking Guidelines

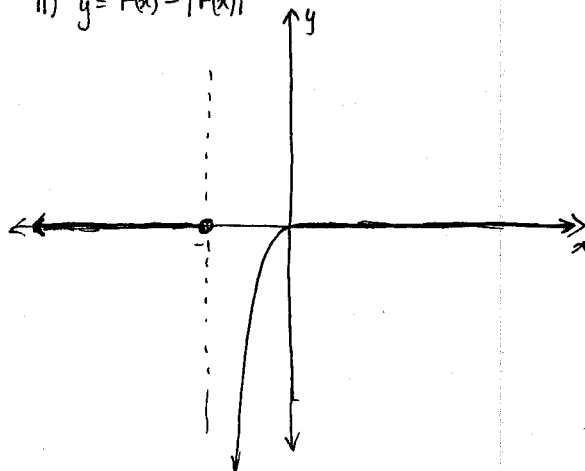
① correct reversal of zero/asymptote

① correct shape

Comments

well done

ii) $y = F(x) - |F(x)|$

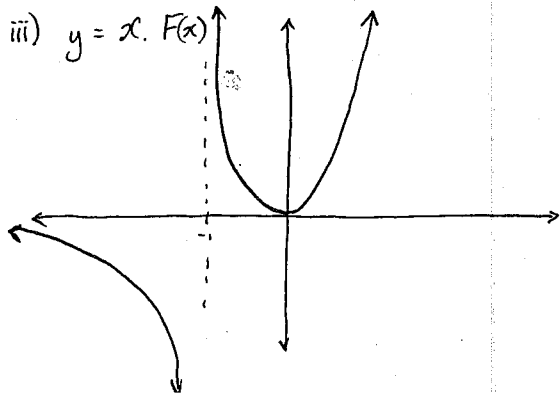


① correct exclusion of asymptote

① zero in correct areas

care needed to show where curve and x-axis co-incide.

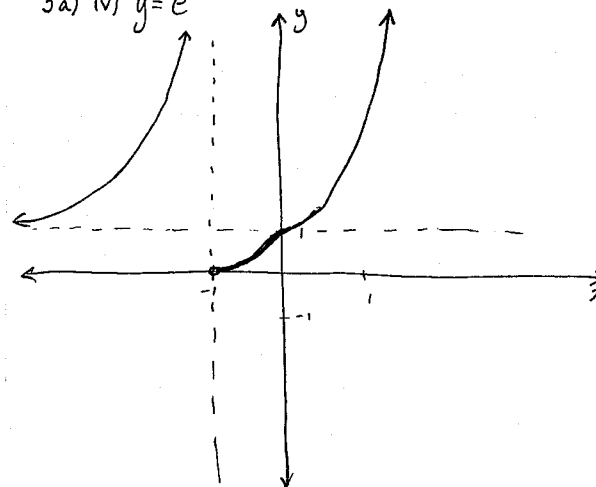
iii) $y = x \cdot F(x)$



① correct asymptote + zeros

① correct shape/regions

3a) iv) $y = e^{F(x)}$



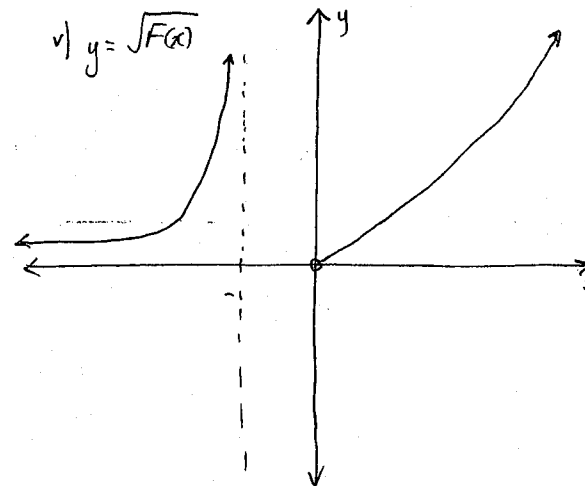
Marking Guidelines

① correct behaviour about $x = 0$

① correct behaviour at $x = -1$

Comments

v) $y = \sqrt{F(x)}$

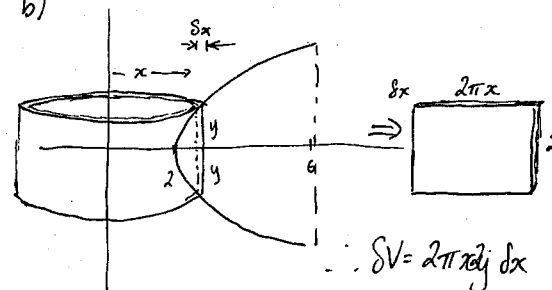


① correct region where $\sqrt{\quad}$ doesn't exist

① correct shape.

care needed when showing where $\sqrt{\quad}$ doesn't exist.

b)



① set up problem with correct δV . (must have some sort of relevant diagram!)

3 b) (cont)

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=2}^6 4\pi y \delta x$$

$$= 4\pi \int_2^6 xy \, dx$$

$$= 4\pi \int_2^6 2x\sqrt{x-2} \, dx$$

using $u = x-2 : du = dx$ and $x = u+2$
 $x=6, u=4 : x=2, u=0$

$$\therefore V = 8\pi \int_0^4 (u+2)u \, du$$

$$= 8\pi \int_0^4 u^{\frac{3}{2}} + 2u^{\frac{1}{2}} \, du$$

$$= 8\pi \left[\frac{2}{5} u^{\frac{5}{2}} + \frac{4}{3} u^{\frac{3}{2}} \right]_0^4$$

$$= 8\pi \left[\left(\frac{2}{5} \cdot 4^{\frac{5}{2}} + \frac{4}{3} \cdot 4^{\frac{3}{2}} \right) - 0 \right]$$

$$= \frac{2816\pi}{15} \text{ cubic units.}$$

4) a) i) $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ (by implicit different)

$$\therefore \frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$= -\frac{b^2 x}{a^2 y}$$

at $P(a \cos \theta, b \sin \theta)$:

$$\frac{dy}{dx} = -\frac{b^2 \cdot a \cos \theta}{a^2 \cdot b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

$\therefore$ tangent is:

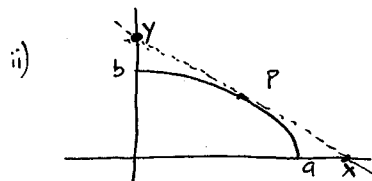
$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$ay \cos \theta - ab \sin^2 \theta = -xb \cos \theta + ab \cos^2 \theta$$

$$bx \cos \theta + ay \sin \theta = ab \cos^2 \theta + ab \sin^2 \theta$$

$\therefore ab \left(\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} \right) = ab (\sin^2 \theta + \cos^2 \theta)$

$$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$



Marking Guidelines

① resolving integral to x 's.

① correct substitutions (incl. limits).

① correct integration.

① correct answer

① correct $\frac{dy}{dx}$ at P .
(inc. differentiation shown)

① correct algebra to tangent eqn.

Comments

many students were unable to complete the integration.

many still unable to reproduce this basic procedure!

many answers not reduced to the standard form.

p7

4) a) (ii) (cont)

at $x: y=0 \Rightarrow \frac{x \cos \theta}{a} = 1$
 $\therefore x = \frac{a}{\cos \theta}$

$\therefore X$ is $(\frac{a}{\cos \theta}, 0)$

at $Y: x=0 \Rightarrow \frac{y \sin \theta}{b} = 1$
 $\therefore y = \frac{b}{\sin \theta}$

$\therefore Y$ is $(0, \frac{b}{\sin \theta})$

iii) $A_{\text{oxy}} = \frac{1}{2} \cdot OX \cdot OY$
 $= \frac{1}{2} \cdot \frac{a}{\cos \theta} \cdot \frac{b}{\sin \theta}$
 $= \frac{ab}{2 \sin \theta \cos \theta}$
 $= \frac{ab}{\sin 2\theta}$

$$\therefore \frac{dA}{d\theta} = -1 \cdot ab (\sin 2\theta)^{-2} \cdot 2 \cos 2\theta$$

$$= \frac{-2ab \cos 2\theta}{\sin^2 2\theta}$$

for $\frac{dA}{d\theta} = 0 \Rightarrow -2ab \cos 2\theta = 0$

or $\cos 2\theta = 0$
 $2\theta = \frac{\pi}{2}$ (1st Quad)
 $\therefore \theta = \frac{\pi}{4}$

test for min: $\frac{d^2A}{d\theta^2} = \frac{+4ab \sin 2\theta}{\sin^3 2\theta} + \frac{-2ab \cos 2\theta \cdot 2 \cos 2\theta}{\sin^3 2\theta}$

$$= \frac{+4ab \sin^2 2\theta + 8ab \cos^2 2\theta}{\sin^3 2\theta}$$

when $\theta = \frac{\pi}{4}$: $\frac{d^2A}{d\theta^2} = \frac{+4ab \sin^2(\frac{\pi}{2}) + 8ab \cos^2(\frac{\pi}{2})}{\sin^3(\frac{\pi}{2})}$
 $= 4ab$
 $> 0 \Rightarrow \text{rel. min.}$

$\therefore \text{Area}_{\min} = \frac{ab}{\sin(\frac{\pi}{2})}$
 $= ab \text{ sq. units}$

when P is $(a \cos \frac{\pi}{4}, b \sin \frac{\pi}{4})$
 or $(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}})$

Marking Guidelines

① finding X

① finding Y .

① expression for area

① expression for $\frac{dA}{d\theta}$

① correct value of $\theta = \frac{\pi}{4}$

① showing rel. min.

① co-ords of P .

Comments

generally well done

no-one showed the area a minimum! This is basic min/max proceed

many missed this part out.

p8

b) i) $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \cdot \tan^2 x \, dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2} x (\sec^2 x - 1) \, dx$$

$$= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} \tan^{n-2} x \, dx$$

$$= \left[\frac{1}{n-1} \tan^{n-1} x \right]_0^{\frac{\pi}{4}} - I_{n-2}$$

$$= \left(\frac{1}{n-1} \tan^{n-1} \frac{\pi}{4} - \frac{1}{n-1} \cdot 0 \right) - I_{n-2}$$

$$= \frac{1}{n-1} - I_{n-2}$$

Marking Guidelines

Comments

① correct $\tan^n x$ method

• most could start this procedure.

① correct integrals

• many tried "by part" at this point! Why?

① correct substitutions

① recognises I_{n-2} .

ii) from $I_n = \frac{1}{n-1} - I_{n-2}$

then $I_5 = \frac{1}{4} - I_3$

$$= \frac{1}{4} - \left(\frac{1}{2} - I_1 \right)$$

now $I_1 = \int_0^{\frac{\pi}{4}} \tan x \, dx$

$$= [\ln(\cos x)]_0^{\frac{\pi}{4}}$$

$$= \ln \frac{1}{\sqrt{2}} - \ln 1$$

$$= \ln 2^{-\frac{1}{2}}$$

$$= -\frac{1}{2} \ln 2$$

$$\therefore I_5 = \frac{1}{4} - \left(\frac{1}{2} - \frac{1}{2} \ln 2 \right)$$

$$= \frac{1}{4} + \frac{1}{2} \ln 2$$

$$= \frac{1}{2} (1 + \ln 2)$$

① correct evaluation of I_1 • many did not use reduction formula properly - tried to find I_3 .

① correct use of reduction formula for answer.

• Evaluate I_1 or I_0 in a reduction formula!• many not able to resolve $\ln \frac{1}{\sqrt{2}}$ to its $\ln 2$ equivalent.

5) a) $Q(x) = x^6 - 3x^2 + 2$

$$Q'(x) = 6x^5 - 6x$$

$$= 6x(x^4 - 1)$$

roots of $Q'(x)$ are $0, \pm 1, \pm i$

$$Q(0) \neq 0 \quad Q(1) = 0 \quad Q(i) \neq 0$$

$$Q(-1) = 0 \quad Q(-i) \neq 0$$

 $\therefore \pm 1$ are the double roots.

① double roots clearly identified (with reasons).

5) a) (cont).

$$\therefore Q(x) = (x-1)^2(x+1)^2 R(x)$$

$$= (x^2 - 2x + 1)(x^2 + 2x + 1) R(x)$$

$$= (x^4 - 2x^2 + 1) R(x)$$

$$\therefore R(x) = \frac{Q(x)}{(x^4 - 2x^2 + 1)}$$

$$\begin{array}{r} x^2 + 2 \\ x^4 - 2x^2 + 1 \overline{) x^6 - 3x^2 + 2} \end{array}$$

$$x^6 - 2x^4 + x^2$$

$$- 2x^4 - 4x^2 + 2$$

$$2x^4 - 4x^2 + 2$$

$$\therefore Q(x) = (x-1)^2(x+1)^2(x^2+2)$$

$$= (x-1)^2(x+1)^2(x+\sqrt{2}i)(x-\sqrt{2}i)$$

① factored over complex field

b) i) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$

now $\alpha + \beta + \gamma = 0$

$$\alpha\beta + \beta\gamma + \alpha\gamma = p$$

$$\alpha\beta\gamma = -1$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = 0^2 - 2(p)$$

$$= -2p$$

since α, β and γ are non-zero and real

$$-2p > 0, \text{ so}$$

$$p < 0$$

for $\alpha^4 + \beta^4 + \gamma^4$: $x^3 = -px - 1$

so $x^4 = -px^2 - x$

ie $\alpha^4 = -p\alpha^2 - \alpha$

$$\beta^4 = -p\beta^2 - \beta$$

$$\gamma^4 = -p\gamma^2 - \gamma$$

$$\therefore \alpha^4 + \beta^4 + \gamma^4 = -p(\alpha^2 + \beta^2 + \gamma^2) - (\alpha + \beta + \gamma)$$

$$= -p(-2p) - 0$$

$$= 2p^2$$

① correct value of $\alpha^2 + \beta^2 + \gamma^2$ ① reasoning for $p < 0$ ① method for x^4 .① correct value of $\alpha^4 + \beta^4 + \gamma^4$

• well done

• well done except for careless errors

5) b) (ii)

for roots $\frac{\alpha}{\beta\gamma}, \frac{\beta}{\alpha\gamma}, \frac{\gamma}{\alpha\beta}$, noting
 $\alpha\beta\gamma = -1$ gives $\frac{\alpha}{\beta\gamma} = \frac{\alpha^2}{\alpha\beta\gamma} = -\alpha^2$

$$\text{i.e. } x = -\alpha^2$$

$$\text{so } \alpha = \sqrt{-x}$$

in original equation:

$$(\sqrt{-x})^3 + p\sqrt{-x} + 1 = 0$$

$$\sqrt{-x}(p-x) = -1$$

$$\text{squaring: } -x(p-x)^2 = 1$$

$$-x(p^2 - 2px + x^2) = 1$$

$$-p^2x + 2px^2 - x^3 = 1$$

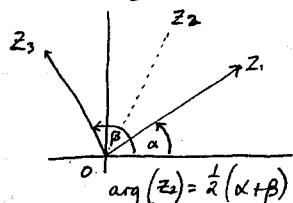
$$\text{so } 0 = x^3 - 2px^2 + p^2x + 1$$

is required eqn.

$$\text{c) } z_1 \cdot z_3 = (z_2)^2$$

$$\text{i.e. } \arg(z_1) + \arg(z_3) = 2\arg(z_2)$$

$$\text{i.e. } \arg(z_2) = \frac{1}{2}(\arg(z_1) + \arg(z_3))$$



$$\arg(z_2) = \frac{1}{2}(\alpha + \beta)$$

$$\text{a) } \arg(z_3) - \arg(z_1) = \arg(z_3) - \arg(z_2)$$

$$\text{a) } \angle z_3 O z_1 = \angle z_3 O z_2$$

$$\Rightarrow O z_2 \text{ bisects } \angle z_3 O z_1$$

Marking Guidelines

① use of $\alpha\beta\gamma = -1$

① using α to form eqn.

① working, including squaring to resolve $\sqrt{-x}$

① final eqn.

① correct arg relationships

① expression for $\arg z_2$

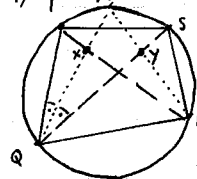
① diagram showing relationships is correct

① correct interpretation linking α, β with args

Comments

• not well done - many students have difficulty connecting complex numbers with geometry.

6) a) i) p



ii) Prove XQRY a cyclic quadrilateral:

$\angle PQM = \angle PRM$ (angles at circumference standing on arc PM)

$\angle PQM = \angle SQM$ (given QM bisects $\angle PQS$)

$\therefore \angle SQM = \angle PRM$ (both = $\angle PQM$)

$\therefore$ 'arc XY' subtends equal angles at Q and R

i.e. $\angle XQY (= \angle SQM) = \angle XRY (= \angle PRM)$

$\therefore$ XQRY is a cyclic quadrilateral.

iii) $\angle RQY = \angle RXY$ (angles at circumference standing on arc RY)

$\angle RPS = \angle RQS$ (angles at circumference standing on arc RS)

$\therefore \angle RXY = \angle RPS$ (both equal to $\angle RQY (= \angle RQS)$)

$\therefore XY \parallel PS$ (corresponding angles are equal)

$$\text{b) } \frac{1}{5} + \frac{2}{5^2} + \frac{3}{5^3} + \dots$$

$$= \frac{1}{5} + \left(\frac{1}{5^2} + \frac{1}{5^2}\right) + \left(\frac{1}{5^3} + \frac{1}{5^3} + \frac{1}{5^3}\right) + \dots$$

$$= \left(\frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \dots\right) + \left(\frac{1}{5^2} + \frac{1}{5^3} + \dots\right) + \left(\frac{1}{5^3} + \frac{1}{5^4} + \dots\right)$$

each is a GP with $r = \frac{1}{5}$, and differing a.s.

i.e. (with $S_0 = \frac{a}{1-r}$):

$$= \frac{\frac{1}{5}}{1-\frac{1}{5}} + \frac{\frac{1}{5^2}}{1-\frac{1}{5}} + \frac{\frac{1}{5^3}}{1-\frac{1}{5}} + \dots$$

$$= \frac{\frac{1}{5}}{\frac{4}{5}} + \frac{\frac{1}{5^2}}{\frac{4}{5}} + \frac{\frac{1}{5^3}}{\frac{4}{5}} + \dots$$

$$= \frac{1}{4} + \frac{1}{20} + \frac{1}{100} + \dots$$

another GP with $a = \frac{1}{4}$, $r = \frac{1}{5}$

Marking Guidelines

① diagram correct.

① linking $\angle Q$ to $\angle R$

① using bisector

① recognising subtended angles on arc XY.

①

① angles from both cyclic quads

① linking these angles

① conclusion (with reason)

① rearranging pattern

① 1st GP groups.

Comments

• poorly done
• diagrams were generally not neat

• most had trouble seeing any pattern

6. b) (cont)

$$\therefore S_{\infty} = \frac{\frac{1}{4}}{\frac{4}{5}} = \frac{5}{16}.$$

"

6) -c) Let $x = \cos \theta$

$$\therefore 16 \cos^4 \theta - 16 \cos^2 \theta + 1 = 0$$

$$\text{or } 8 \cos^4 \theta - 8 \cos^2 \theta + 1 = \frac{1}{2}$$

$$\text{or } \cos 4\theta = \frac{1}{2}$$

$$\therefore 4\theta = 2n\pi \pm \cos^{-1}\left(\frac{1}{2}\right) \\ = 2n\pi \pm \frac{\pi}{3} \\ = \frac{\pi(6n \pm 1)}{3}$$

$$\therefore \theta = \left(\frac{6n \pm 1}{12}\right)\pi \quad n=0, \pm 1, \pm 2$$

$$\therefore \text{for } 0 \leq \theta \leq 2\pi : \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}, \frac{23\pi}{12}$$

for $x = \cos \theta$; unique values are

$$x_1 = \cos \frac{\pi}{12} \quad (= \cos \frac{23\pi}{12})$$

$$x_2 = \cos \frac{5\pi}{12} \quad (= \cos \frac{19\pi}{12})$$

$$x_3 = \cos \frac{7\pi}{12} \quad (= \cos \frac{17\pi}{12})$$

$$x_4 = \cos \frac{11\pi}{12} \quad (= \cos \frac{13\pi}{12})$$

$$\text{we also note } \cos \frac{\pi}{12} = -\cos \frac{11\pi}{12} \quad (x_1 = -x_4)$$

$$\text{and } \cos \frac{5\pi}{12} = -\cos \frac{7\pi}{12} \quad (x_2 = -x_3)$$

Similarly, by using $u = x^2$, we get

$$16u^2 - 16u + 1 = 0$$

$$\text{ie } u = \frac{16 \pm \sqrt{192}}{32}$$

$$= \frac{16 \pm 4\sqrt{3}}{32}$$

$$= \frac{1}{4}(2 \pm \sqrt{3})$$

$$\therefore x^2 = \frac{1}{4}(2 \pm \sqrt{3})$$

$$\text{so } x = \pm \sqrt{\frac{1}{4}(2 \pm \sqrt{3})}$$

These must correspond to the x_1 to x_4 above, so $\cos \frac{\pi}{12}$ is the largest positive, then:

$$\cos \frac{\pi}{12} = \frac{1}{2}\sqrt{2 + \sqrt{3}}$$

and $\cos \frac{5\pi}{12} = \frac{1}{2}\sqrt{2 - \sqrt{3}}$ (the other one)

Marking Guidelines

① 2nd GP with correct soln.

① values for θ - generator or other method.

① unique values for roots

① roots in terms of surds

① $\cos \frac{\pi}{12}$ correct① $\cos \frac{5\pi}{12}$ correct

(both with reason).

Comments

this is a fairly standard question - poorly done (no matter which method was used).

many had difficulties resolving the unique roots

many could not do this part.

many could not indicate why they made their choice.

7) (a)

i) α, x_1, x_2 and β satisfy $y = x^2(4-x)^2$ for the given y value. Considering y constant;

$$\therefore x^2(4-x)^2 - y = 0$$

$$x^2(16 - 8x + x^2) - y = 0$$

$$\text{or } x^4 - 8x^3 + 16x^2 - y = 0$$

ii) from the graph, $x=2$ is an axis of symmetry, so 2 is the midpoint between x_1 and x_2 , α and β .

$$\therefore \frac{x_1 + x_2}{2} = 2$$

$$\frac{\alpha + \beta}{2} = 2$$

$$\text{ie } x_1 + x_2 = 4$$

$$\alpha + \beta = 4$$

$$\therefore x_1 + x_2 + \alpha + \beta = 8 \quad (\text{from co-eff, as well as above})$$

by 3 roots at a time:

$$x_1 x_2 \alpha + x_1 x_2 \beta + x_1 \alpha \beta + x_2 \alpha \beta = 0$$

$$(\alpha + \beta) x_1 x_2 + (x_1 + x_2) \alpha \beta = 0$$

$$\text{but } \alpha + \beta = 4, \quad x_1 + x_2 = 4 \quad \text{gives}$$

$$4x_1 x_2 + 4\alpha \beta = 0$$

$$\text{ie } \alpha \beta = -x_1 x_2 \quad \text{as reqd.}$$

from 4 roots together:

$$\alpha \beta x_1 x_2 = -y$$

$$\text{ie } x_1 x_2 (-x_1 x_2) = -y$$

$$\text{gives } (x_1 x_2)^2 = y$$

$$\text{or } x_1 x_2 = \sqrt{y}$$

for $x_2 - x_1$, we note

$$(x_2 - x_1)^2 = (x_2 + x_1)^2 - 4x_1 x_2$$

$$= 4^2 - 4\sqrt{y}$$

$$= 16 - 4\sqrt{y}$$

as $x_2 > x_1$,

$$x_2 - x_1 = +\sqrt{16 - 4\sqrt{y}} \\ = 2\sqrt{4 - \sqrt{y}} //$$

Marking Guidelines

① correct deduction + re-arrangement of equation.

① correct explanation

① correct roots

① correct working to result.

① correct working to result

① expression for $(x_2 - x_1)^2$

① correct working to result.

Comments

well done

well done

these steps proved difficult for many students

7(a) iii)

The slice has volume SV , given by

$$SV = \pi(x_2^2 - x_1^2) \delta y$$

$$= \pi(x_2 - x_1)(x_2 + x_1) \delta y$$

$$= 8\pi \sqrt{4-y} \delta y$$

(using results from (ii))

Hence

$$V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^{16} 8\pi \sqrt{4-y} \delta y$$

$$= 8\pi \int_0^{16} \sqrt{4-y} dy$$

$$\text{using } y = (4-u)^2 \quad y=0, \quad u=4$$

$$dy = -2(4-u)du \quad y=16, \quad u=0$$

$$4-y = 4-(4-u)^2$$

$$= u$$

$$\therefore V = 8\pi \int_4^0 \sqrt{u} \cdot (-2(4-u)) du$$

$$= 16\pi \int_4^0 u^{3/2} - 4u^{1/2} du$$

$$= 16\pi \left[\frac{2}{5} u^{5/2} - \frac{8}{3} u^{3/2} \right]_4^0$$

$$= 16\pi \left[0 - \left(\frac{2}{5} 4^{5/2} - \frac{8}{3} 4^{3/2} \right) \right]$$

$$= 16\pi \left(-\frac{64}{5} + \frac{64}{3} \right)$$

$$= \frac{2048\pi}{15} \text{ cubic units.}$$

$$b) i) \text{ from } y = Vt \sin \alpha - \frac{1}{2}gt^2$$

$$y=0: 0 = Vt \sin \alpha - \frac{1}{2}gt^2$$

$$= t(V \sin \alpha - \frac{1}{2}gt)$$

$$\text{ie } t=0 \text{ or } t = V \sin \alpha - \frac{1}{2}gt$$

$$\frac{1}{2}gt = V \sin \alpha$$

$$\text{so } t = \frac{2V \sin \alpha}{g} \text{ as reqd.}$$

Marking Guidelines

① correct SV ① correct working to show V .

① correct limits for sub.

① correct answer.

① working correct to result.

p15

Comments

• most students could establish V as required.

• mostly well done

p16

7) b) ii)

$$\text{max range when } t = \frac{2V \sin \alpha}{g}$$

$$\text{in } x = Vt \cos \alpha = \frac{2V^2 \sin \alpha \cos \alpha}{g} \quad m$$

iii) position of target T is given by

$$x = d + ut$$

$$\text{to hit the target, } t = \frac{2V \sin \alpha}{g} \text{ and } x = \frac{2V^2 \sin \alpha \cos \alpha}{g}$$

substituting in above;

$$\frac{2V^2 \sin \alpha \cos \alpha}{g} = d + u \cdot \frac{2V \sin \alpha}{g}$$

$$\text{ie } u \cdot \frac{2V \sin \alpha}{g} = \frac{2V^2 \sin \alpha \cos \alpha - dg}{g}$$

$$\therefore u = \frac{2V^2 \sin \alpha \cos \alpha - dg}{2V \sin \alpha} = V \cos \alpha - \frac{gd}{2V \sin \alpha}$$

① correct subst.

① initial setup correct

① substitutions correct

① correct working to answer.

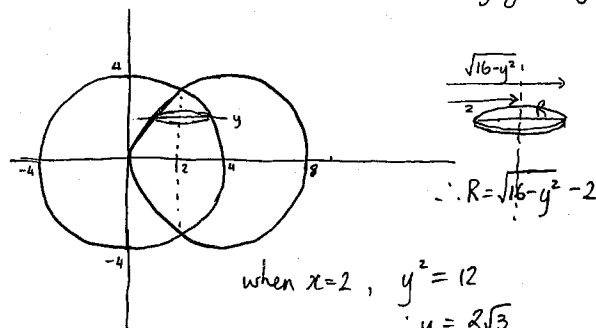
$$8) a) x^2 + y^2 = 16$$

center $(0,0)$

radius 4

$$x^2 - 8x + y^2 = 0$$

$$x^2 - 8x + 16 + y^2 = 16$$

center $(4,0)$ radius 4.common chord is $x=2$ (by symmetry)

$$\text{when } x=2, \quad y^2 = 12$$

$$\therefore y = 2\sqrt{3}$$

for slice at y :

$$SA = \pi (\sqrt{16-y^2} - 2)^2$$

$$= \pi (16 - y^2 - 4\sqrt{16-y^2} + 4)$$

$$= \pi (20 - y^2 - 4\sqrt{16-y^2})$$

① correct circles (including diagram) and common chord

① correct R ① correct y limit① correct expression for SA

• most did not draw a slice/shell diagram so got an incorrect radius expression
• many used shells - very poorly! Shells works best when the axis of rotation is outside the region being rotated.

8) a) (cont)

$$\therefore \delta V = \delta A \cdot \delta y$$

$$= \pi (20 - y^2 - 4\sqrt{16 - y^2}) \delta y$$

$$\therefore V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^{2\sqrt{3}} \pi (20 - y^2 - 4\sqrt{16 - y^2}) \delta y$$

$$= \int_{-2\sqrt{3}}^{2\sqrt{3}} \pi (20 - y^2 - 4\sqrt{16 - y^2}) dy$$

$$= 2\pi \int_0^{2\sqrt{3}} (20 - y^2 - 4\sqrt{16 - y^2}) dy \quad (\text{from symmetry})$$

$$= 2\pi \left[20y - \frac{1}{3}y^3 \right]_0^{2\sqrt{3}} - 2\pi \int_0^{2\sqrt{3}} 4\sqrt{16 - y^2} dy$$

using $y = 4\sin\theta$ $y=0 \Rightarrow \theta=0$
 $dy = 4\cos\theta d\theta$ $y=2\sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$

$$= 2\pi \left[40\sqrt{3} - \frac{1}{3}(2\sqrt{3})^3 - (0) \right] - 8\pi \int_0^{\frac{\pi}{3}} \sqrt{16 - 16\sin^2\theta} \cdot 4\cos\theta d\theta$$

$$= 2\pi (40\sqrt{3} - 8\sqrt{3}) - 128\pi \int_0^{\frac{\pi}{3}} \cos^2\theta d\theta$$

$$= 64\pi\sqrt{3} - 128\pi \int_0^{\frac{\pi}{3}} \frac{1}{2}(1 + \cos 2\theta) d\theta$$

$$= 64\pi\sqrt{3} - 64\pi \left[\theta + \frac{1}{2}\sin 2\theta \right]_0^{\frac{\pi}{3}}$$

$$= 64\pi \left(\sqrt{3} - \left[\frac{\pi}{3} + \frac{1}{2} \cdot \frac{\sqrt{3}}{2} - 0 \right] \right)$$

$$= 64\pi \left(\frac{3\sqrt{3}}{4} - \frac{\pi}{3} \right) \text{ cu. units } "$$

b) ii) for H: $b^2 = a^2(e^2 - 1)$

$$e^2 - 1 = \frac{b^2}{a^2}$$

$$\text{so } e^2 = \frac{b^2}{a^2} + 1$$

$$= \frac{b^2 + a^2}{a^2}$$

For E; call its eccentricity ϵ

$$\therefore b^2 = (a^2 + b^2)(1 - \epsilon^2)$$

$$\text{u } 1 - \epsilon^2 = \frac{b^2}{a^2 + b^2}$$

$$\text{or } \epsilon^2 = 1 - \frac{b^2}{a^2 + b^2}$$

Marking Guidelines

Comments

① correct expression for V

those using shells set up the wrong radius expression with the wrong 'y' expression and usually the wrong limits!

① working to substitution

① correctly working in substitution

① correct answer

use a different symbol for the eccentricity of the ellipse!

8) b) (i) (cont)

$$\therefore \epsilon^2 = \frac{(a^2 + b^2) - b^2}{a^2 + b^2}$$

$$= \frac{a^2}{a^2 + b^2}$$

$$= \frac{1}{e^2}$$

$$\therefore \epsilon = \frac{1}{e}$$

hence the ellipse E has eccentricity $\frac{1}{e}$,

ii) For P in quadrant I;

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{--- ①} \quad \frac{x^2}{(a^2 + b^2)} + \frac{y^2}{b^2} = 1$$

but from (i) $e^2 = \frac{a^2 + b^2}{a^2}$

$$\text{so } a^2 + b^2 = a^2 e^2$$

$$\text{giving } \frac{x^2}{a^2 e^2} + \frac{y^2}{b^2} = 1 \quad \text{--- ②}$$

① + ② gives: $\frac{x^2}{a^2 e^2} + \frac{x^2}{a^2} = 2$

$$\frac{x^2}{a^2} (e^2 + 1) = 2$$

$$\text{or } \frac{x^2}{a^2 e^2} (1 + e^2) = 2$$

$$e^2 \times \text{②}: \frac{x^2}{a^2} + \frac{y^2}{b^2} = e^2 \quad \text{--- ③}$$

so ③ - ① gives: $\frac{e^2 y^2}{b^2} + \frac{y^2}{b^2} = e^2 - 1$

$$\frac{y^2}{b^2} (e^2 + 1) = e^2 - 1$$

$$\therefore \text{at P } x^2 = \frac{2a^2 e^2}{e^2 + 1} \quad y^2 = \frac{b^2 (e^2 - 1)}{e^2 + 1}$$

$$\text{so } x = ae \sqrt{\frac{2}{e^2 + 1}} \quad \text{so } y = b \sqrt{\frac{e^2 - 1}{e^2 + 1}}$$

for tan α , we need the angle

$$\text{at P on H: } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\text{at P: } \frac{dy}{dx} = \frac{b^2}{a^2} \cdot \frac{ae \sqrt{2}}{\sqrt{e^2 + 1}} \cdot \frac{\sqrt{e^2 - 1}}{b \sqrt{e^2 + 1}}$$

$$= \frac{b}{a} e \sqrt{2} \cdot \frac{1}{\sqrt{e^2 - 1}}$$

for H: $b^2 = a^2(e^2 - 1)$

$$\text{or } b = a \sqrt{e^2 - 1}$$

$$\text{so } \frac{dy}{dx} = \frac{a \sqrt{e^2 - 1}}{a} e \sqrt{2} \cdot \frac{1}{\sqrt{e^2 - 1}}$$

$$= e \sqrt{2}$$

at P on E:

$$\frac{x^2}{a^2 + b^2} + \frac{y^2}{b^2} = 1$$

$$\text{or } \frac{x^2}{a^2 e^2} + \frac{y^2}{b^2} = 1$$

Marking Guidelines

Comments

① working to $\frac{1}{e}$.

① working for simultaneous eqns.

① correct values of x, y at P

① correct use of $b^2 = a^2(e^2 - 1)$

① tangent gradient at P on H.

most did not find the point P - or tried to give it generic co-ordinates.

most who got this far did not use this to get expressions in terms of e .

8) b) ii) (cont)

$$\begin{aligned} \therefore \frac{2x}{a^2 e^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} &= 0 \\ \therefore \frac{dy}{dx} &= -\frac{b^2}{a^2 e^2} \cdot \frac{x}{y} \\ \text{at P: } \frac{dy}{dx} &= -\frac{b^2}{a^2 e^2} \cdot \frac{ae\sqrt{2}}{\frac{ae\sqrt{2}}{\sqrt{e^2+1}}} \cdot \frac{\sqrt{e^2+1}}{b\sqrt{e^2-1}} \\ &= -\frac{b\sqrt{2}}{ae} \cdot \frac{1}{\sqrt{e^2-1}} \\ &= -\frac{\sqrt{2}}{e} \cdot \frac{1}{a\sqrt{e^2-1}} \cdot \frac{1}{\sqrt{e^2-1}} \\ &= -\sqrt{2} \frac{1}{e} \end{aligned}$$

$$\text{ie } m_1 = e\sqrt{2} \quad m_2 = -\sqrt{2} \frac{1}{e}$$

$$\begin{aligned} \therefore \tan \alpha &= \left| \frac{\sqrt{2}e - (-\sqrt{2} \frac{1}{e})}{1 + \sqrt{2}e(-\sqrt{2} \frac{1}{e})} \right| \\ &= \left| \frac{\sqrt{2}(e + \frac{1}{e})}{1 - 2} \right| \\ &= \sqrt{2} \left(e + \frac{1}{e} \right) \end{aligned}$$

Marking Guidelines

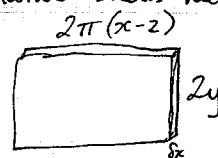
Comments

① tangent gradient at P on E.

① correct working in angle between two lines formula.

• whilst many recognised the need to use this, the lack of co-ords for P in terms of 'e' made this impossible to work out.

Alternative "Shells" method solution to Q8 (a)



$$\delta V = 2\pi \cdot 2y (x-2) \delta x$$

$$= 4\pi y (x-2) \delta x$$

$$= 4\pi (x-2) \sqrt{16-x^2} \delta x$$

$$\therefore V = 4\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (x-2) \sqrt{16-x^2} dx$$

$$= 4\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (4\sin\theta - 2) \sqrt{16-16\sin^2\theta} \cdot 4\cos\theta d\theta$$

$$= 16\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2(\sin\theta - 1) 4\cos\theta \cos\theta d\theta$$

$$= 128\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (2\sin\theta - 1) \cos^2\theta d\theta$$

$$= 128\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2\sin\theta \cos^2\theta - \cos^2\theta d\theta$$

$$= 128\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2\sin\theta \cos^2\theta - \left(\frac{1}{2} + \frac{1}{2}\cos 2\theta\right) d\theta$$

$$= 128\pi \left[-\frac{2}{3}\cos^3\theta - \frac{\theta}{2} - \frac{1}{4}\sin 2\theta \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$= 128\pi \left[0 - \frac{\pi}{4} - 0 - \left(-\frac{2}{3}\left(\frac{\sqrt{3}}{2}\right)^3 - \frac{\pi}{12} - \frac{1}{4}\left(\frac{\sqrt{3}}{2}\right) \right) \right]$$

$$= 128\pi \left[-\frac{\pi}{4} + \frac{\sqrt{3}}{4} + \frac{\pi}{12} + \frac{\sqrt{3}}{8} \right]$$

$$= 128\pi \left[\frac{3\sqrt{3}}{8} - \frac{\pi}{6} \right]$$

$$= 64\pi \left[\frac{3\sqrt{3}}{4} - \frac{\pi}{3} \right]$$

✓✓

• note radius expression if using circle, centered at O.

• note limits, given use of $\uparrow$.

$$\begin{aligned} x &= 4\sin\theta & x=2 & \theta = \frac{\pi}{6} \\ dx &= 4\cos\theta d\theta & x=4 & \theta = \frac{\pi}{2} \end{aligned}$$